

Followup 26-02

Out of hospital PEA arrest the day after a new opioid prescription

Patient	71-year-old female. History of major depressive disorder, GERD, hyperlipidemia, osteoarthritis, prior right total hip arthroplasty, 45 pack-year smoking history (quit approximately eight years prior), regular cannabis use. Seen and discharged from an emergency department the previous day with an intra-articular right distal radius fracture and an ulnar styloid fracture sustained when she was pushed to the ground, treated with hematoma block, closed reduction, and a sugar tong splint, and sent home with a new prescription for hydrocodone-acetaminophen.
Outcome	ROSC after approximately one hour of arrest, hypoxic-ischemic brain injury with myoclonic status epilepticus, transitioned to comfort care on hospital day 3, died early on hospital day 4.

The call

Dispatch was for nausea and vomiting. The patient was conscious when first responders arrived. She arrested in front of them.

The transporting ALS crew arrived to find manual CPR in progress with bag valve mask ventilation, an advanced EMT with an IO established and fluid running, and an i-gel already placed with chest rise and fall, positive lung sounds, negative epigastric sounds, and a positive waveform capnography tracing. Family reported the wrist fracture from the day before and no other significant medical history. No medication list was obtainable on scene.

Field course, relative to ALS arrival:

- Initial rhythm PEA. Manual CPR transitioned to mechanical compressions.
- Naloxone 2 mg IO.
- Epinephrine 1:10,000, 1 mg IO, two doses.
- ROSC into an organized sinus rhythm. 12 lead showed sinus rhythm with no ST segment abnormality.
- First post-ROSC vitals: 124/68, MAP 87, pulse 93. **Blood glucose 349.**
- Blood pressure then fell and stayed down: 74/45 (MAP 55), 71/45 (MAP 54), 67/42 (MAP 50), 70/46 (MAP 54), 72/46 (MAP 55).
- Push dose epinephrine 1:100,000, 20 mcg IO, five doses spread across the remainder of the call.
- Normal saline 1,500 mL IO total.
- Peripheral IV established en route.
- Alert given to the receiving facility. No orders requested or received.
- Late in transport: 128/60 (MAP 83), then 106/54 (MAP 71), SpO2 99%. Spontaneous respiratory effort returned with grunting respirations. Ventilation continued to be assisted with BVM at 100% oxygen. No purposeful movement, no command following.

The number to hold onto for later: **blood glucose 349**.

Primary impression

Cardiac arrest, PEA, etiology undetermined, with recurrent post-ROSC hypotension.

Correct, and correctly limited. Nothing available in the back of the truck identified a cause, and the hospital, with a CT scanner, an echocardiogram, blood cultures, and a send-out toxicology panel, did not name one either. The final record attributes death to anoxic brain injury secondary to cardiac arrest and leaves the arrest itself undifferentiated.

Three pieces of information the crew documented turned out to be the most useful data in the entire chart: the vomiting that preceded the arrest, the pattern of the hypotension, and the glucose.

Hospital course

Emergency department. She arrived comatose with spontaneous respiratory effort, hypotensive, and unable to protect her airway. The field i-gel was exchanged for a 7.0 cuffed endotracheal tube on a single video-assisted attempt, confirmed by direct visualization, waveform capnography, chest rise, bilateral breath sounds, and chest radiograph. Shock worsened and she was started on norepinephrine, then epinephrine infusion. Vancomycin and piperacillin-tazobactam were started for suspected aspiration given the vomiting, the leukocytosis, and post-arrest shock. A broad search for a reversible cause was launched: CT of the head, CT of the thorax, abdomen and pelvis, blood cultures, respiratory viral panel, serum and urine toxicology including alcohols and acetaminophen, and serial high-sensitivity troponin.

CT of the head showed no acute intracranial abnormality, no hemorrhage, and no acute infarct, with mild chronic volume loss, chronic microvascular disease, and an old left frontal periventricular ischemic infarct. CT of the thorax, abdomen and pelvis excluded aortic dissection, central pulmonary embolism, pneumothorax, and intra-abdominal traumatic injury. It showed bibasilar and right middle lobe atelectasis, secretions in the right mainstem bronchus and bronchus intermedius, and bilateral rib fractures attributable to CPR. The endotracheal tube was noted to abut the carina and was withdrawn.

Hospital day 1. She remained comatose. Echocardiogram showed an ejection fraction of 57% with no apparent wall motion abnormality. The epinephrine infusion was discontinued early in the day and norepinephrine was weaned off by mid-morning, with a stated ICU target of MAP greater than 65. MRSA nares PCR returned negative and vancomycin was stopped; piperacillin-tazobactam was narrowed to ampicillin-sulbactam with a planned five day course for empiric aspiration coverage. Myoclonic jerking was observed and continuous video EEG was started. It showed myoclonic status epilepticus from the onset of recording until approximately mid-afternoon, transitioning to burst suppression with identical bursts. She was treated with propofol, levetiracetam, and fosphenytoin, and the myoclonus resolved. Neurology documented persistent coma, post-arrest myoclonus, concern for nonconvulsive status epilepticus, and a very poor prognosis for independent functional recovery, and recommended MRI. High-sensitivity troponin peaked and creatine kinase returned mildly elevated. Send-out urine toxicology resulted.

Hospital day 2. Continuous EEG after propofol was discontinued showed persistent burst suppression with identical bursts and no further seizures. Both EEG reports interpreted the pattern as highly suggestive of severe

anoxic brain injury with a grave prognosis for functional neurologic recovery. Blood cultures showed no growth. Ventilator support continued at a rate of 20, tidal volume 470 mL, PEEP 5, FiO₂ 30%. Hypertension emerged and was attributed to autonomic dysfunction following anoxic brain injury, with PRN labetalol ordered.

Hospital day 3. Arterial blood gas on FiO₂ 30% showed adequate ventilation and gas exchange. She remained unable to protect her airway after sedation was weaned and showed no purposeful response. Goals of care were discussed with her husband and family, comfort measures were elected, and she was extubated and transferred out of the intensive care unit. Care transitioned to inpatient hospice.

Hospital day 4. She died in the early morning hours.

Final diagnoses: out of hospital PEA cardiac arrest of undetermined etiology; hypoxic-ischemic (anoxic) brain injury; post-cardiac-arrest myoclonic status epilepticus; undifferentiated shock, resolved; suspected aspiration; acute right distal radius and ulnar styloid fractures; bilateral CPR-associated rib fractures.

Education points

Oxygen has no reservoir

Nearly every substrate the body needs is stored somewhere: glycogen in liver and muscle, fat in adipose tissue, calcium in bone, iron in ferritin. Oxygen is the exception. The only oxygen a person has is what is dissolved in plasma, bound to hemoglobin in circulating blood, and sitting in the functional residual capacity of the lungs. Added together in a healthy adult that is roughly one and a half liters, against a resting consumption of about 250 mL per minute. That is the entire margin, and it is measured in minutes.

This is why ventilation problems kill faster than almost anything else, and why the sequence here moved as fast as it did. The record describes nausea and vomiting, then respiratory distress, then pulselessness. That is not a cardiac arrest that happened to involve breathing. That is a respiratory arrest that ran out of margin.

The path from hypoventilation to PEA

When ventilation slows or stops while circulation continues, two things happen in parallel and they are not the same thing.

Carbon dioxide accumulates. CO₂ is produced continuously by metabolism and cleared only by alveolar ventilation, so if minute ventilation halves, arterial CO₂ roughly doubles. CO₂ combines with water to form carbonic acid, which dissociates into hydrogen ion and bicarbonate, so a rising CO₂ is a falling pH by definition.

Oxygen falls at the same time. Alveolar oxygen is determined in part by alveolar CO₂, so as CO₂ climbs the space available for oxygen in the alveolus shrinks even on room air. Hypoventilation causes hypoxia and hypercarbia together, and the hypoxia is what does the damage.

Both act on the myocardium. Cardiac cells need oxygen to make the ATP that runs the sodium-potassium pump and the calcium handling machinery of contraction, and acidosis reduces the sensitivity of the contractile proteins to calcium. Contractility falls, then blood pressure, then coronary perfusion, which reduces myocardial oxygen delivery further. The conduction system is more resistant to hypoxia than the contractile machinery, and that is the key point: **the heart keeps generating organized electrical activity after it has stopped generating meaningful mechanical output.** That gap is what a monitor displays as PEA.

So PEA is not really a rhythm. Ventricular fibrillation is a rhythm problem with an electrical solution. PEA is a plumbing or metabolic problem being reported by an electrical instrument. The monitor is telling you the conduction system is still alive and asking what killed everything else.

Why the brain is the organ that decides the outcome

The brain is about 2% of body weight and consumes about 20% of resting oxygen delivery. It stores essentially no glycogen and cannot generate meaningful ATP anaerobically for more than a few minutes. When ATP fails, the sodium-potassium ATPase stops, membrane potential collapses, and neurons depolarize. Depolarization dumps glutamate into the synapse. Glutamate opens channels that let calcium into the cell. Intracellular calcium activates proteases, lipases, and nitric oxide synthase, which begin taking the cell apart from the inside. This is excitotoxicity, and it continues after the oxygen comes back, because the enzymes are already switched on.

That is the reason this patient's kidneys, liver, and heart all recovered and she did not. Creatinine stayed at 0.58. Ejection fraction was 57%. Transaminases were only mildly up. Roughly one hour of arrest, and every organ except the brain walked away from it. Restoring perfusion is not the same as restoring function, and the organ with the least tolerance sets the ceiling on the outcome from the first minutes onward.

Further reading on this paragraph. The cascade above is dense, and it takes some more studying to understand well. Two resources cover it properly.

The same cellular sequence, ATP failure to pump failure to depolarization to glutamate to calcium to cell death, is what plays out in ischemic stroke, and Ninja Nerd draws the whole thing out visually in Acute Ischemic Stroke, free on YouTube: <https://www.youtube.com/watch?v=abxcrAvw-O0>

For the cardiac arrest version specifically, including why a second wave of injury arrives after ROSC and how hyperoxia, CO₂, temperature, and perfusion pressure feed into it, Sekhon and colleagues published an open access review in Critical Care under the title Clinical pathophysiology of hypoxic ischemic brain injury after cardiac arrest: a two-hit model. It is free to read in full: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5390465/>

Post-cardiac-arrest syndrome

ROSC is not the end of the disease. It is the start of a different one, with four parts that run at the same time:

1. **Brain injury.** Excitotoxicity, cerebral edema, disturbed autoregulation, and loss of the normal coupling between cerebral blood flow and metabolic demand.
2. **Myocardial dysfunction.** Global stunning of the myocardium. It is usually reversible over 24 to 48 hours, and in this patient it was.
3. **Systemic ischemia and reperfusion response.** Whole-body ischemia followed by reperfusion produces a cytokine surge, endothelial activation, and vasoplegia that closely resembles sepsis. This is why the record says undifferentiated shock and why she needed pressors while her heart was pumping fine.
4. **Persistence of whatever caused the arrest.**

Item three explains what the crew saw. They gave 1,500 mL and five doses of push dose epinephrine and watched the pressure fall back each time. The crew was not doing anything wrong. Post-arrest vasoplegia was reasserting itself against every intervention aimed at it, and it kept doing so in the emergency department until a norepinephrine infusion was running.

Aspiration, and why the right side

She was vomiting before she arrested and CT found secretions in the right mainstem bronchus and bronchus intermedius, with right middle lobe and bibasilar atelectasis.

The trachea divides at the carina into two mainstem bronchi that are not symmetric. The right mainstem is shorter, wider, and leaves the trachea at a shallower angle, roughly 25 degrees off vertical compared with about 45 degrees on the left. Anything that falls down the trachea, gastric contents, a tooth, an endotracheal tube advanced too far, preferentially goes right. Within the right lung, which segments fill depends on the position of the patient. Upright, it is the lower lobe. Supine, which is how most of these patients are found and how all of them are transported, it is the posterior segment of the upper lobe and the superior segment of the lower lobe. The right middle lobe involvement here fits a patient who was down, vomiting, and then supine on a cot.

Two distinct things follow aspiration and they get confused constantly. **Aspiration pneumonitis** is a chemical burn. Gastric acid injures alveolar epithelium and capillary endothelium within minutes, producing inflammation, exudate, and infiltrates on imaging with no organism involved. **Aspiration pneumonia** is an actual bacterial infection with oropharyngeal flora, and it takes a day or more to establish. Antibiotics do nothing for the first. That distinction is exactly why this patient's antibiotic course was narrowed and time-limited rather than escalated, and it is why the imaging showing atelectasis and secretions rather than consolidation mattered.

Post-arrest myoclonus and status epilepticus

Anoxic neurons do not simply go quiet. Injured cortex loses its inhibitory interneurons preferentially, because those cells are among the most metabolically demanding and die first. What is left is excitatory tissue with the brakes removed, and it fires in synchronized bursts. Clinically that shows as myoclonus, sudden shock-like jerks. On EEG it can meet criteria for status epilepticus even when the movements are subtle or absent, which is why continuous EEG was started rather than relying on watching her.

Burst suppression is a pattern of high-voltage bursts of activity separated by near-flat intervals. It can be produced deliberately with anesthetic, and propofol will do it. It can also be the signature of a cortex so damaged it can only generate intermittent synchronized discharges. Telling the two apart is the whole game in post-arrest neuroprognostication, and it is why the second EEG, recorded after propofol was discontinued, carried more weight than the first. The pattern persisted without the drug.

The detail that both reports flagged is **identical bursts**: each burst looking essentially the same as the last, stereotyped and unvarying. Normal cortex does not produce identical output. Variability requires a functioning network. Identical bursts after cardiac arrest are strongly associated with severe hypoxic-ischemic injury and poor neurologic outcome, and that is the specific finding that turned "very poor prognosis" on day 1 into "grave prognosis" on day 2.

Prehospital considerations

The physiology above reduces to a small number of things a crew actually controls.

Ventilation is the intervention with the shortest fuse. In a hypoventilating patient the entire problem is minute ventilation, and it is fixable with a BVM before it becomes an arrest. The most valuable minutes in this patient's course were before the arrest, not after it.

In PEA, the rhythm is not the target. Compressions and the cause are the target. Everything else is supportive.

Compression quality is coronary perfusion pressure. Aortic diastolic pressure minus right atrial pressure is what pushes blood into coronary arteries, and it only exists during the recoil phase. Every pause bleeds it back to zero and it takes several compressions to rebuild. That is the mechanism behind every rule about minimizing interruptions.

Post-ROSC hypotension is a warning about the brain. Cerebral autoregulation is broken after arrest, which means cerebral blood flow becomes directly dependent on MAP. This crew's documented MAPs of 50 to 55 are the same number the ICU spent day 0 and day 1 chasing with infusions, with an explicit target of greater than 65. The pressure the crew is fighting for is brain perfusion.

A saturation of 94 to 100 is where you want this patient, and 100 is the top of a scale rather than a measurement of oxygen. Pulse oximetry reports the percentage of hemoglobin binding sites that are occupied. Once essentially all of them are full the number stops moving, and it stops moving long before oxygen stops accumulating. Two patients can both read 100%: one with an arterial oxygen tension of 90 mmHg and one at 400. The oximeter cannot tell them apart, because hemoglobin has no more room in either case and the entire difference is oxygen dissolved in plasma, which the oximeter does not see. So target a high saturation, and understand that once the number pins at 100 it has stopped telling you how much oxygen is actually circulating.

Oxygen at the extremes is not inert. When ischemic tissue is abruptly reperfused, some of the arriving oxygen is converted into **free radicals**, highly reactive molecules that oxidize cell membranes, proteins, and DNA. Cells normally neutralize these with a set of dedicated enzymes, but a prolonged ischemic period depletes those defenses, so the tissue is least equipped to handle radicals at the moment it is generating the most of them. More oxygen means more raw material for that reaction, and it extends the injury the arrest started.

The tradeoff is worst at the top of the curve. Once hemoglobin is saturated, additional oxygen adds almost nothing to delivery, because the extra is dissolved in plasma and plasma carries very little of it. So above saturation you are buying essentially no delivery and paying in free radicals. High oxygen tension also constricts cerebral arterioles, as does the low CO₂ that comes from bagging too fast, and both reduce the blood flow you are trying to protect.

This patient was documented at SpO₂ 99% on BVM with 100% oxygen, which is a reasonable place to be. It is also a place where the monitor has stopped telling you what dose you are giving.

The interventions

PREHOSPITAL INTERVENTIONS

Prehospital airway management

By the time the ALS crew arrived, an i-gel was in and it was working. Two separate good decisions are stacked in that one sentence, and neither one is automatic.

The first belongs to the limited ALS crew. They placed an appropriately sized supraglottic airway in an arrested patient and then confirmed it the right way: chest rise and fall, positive lung sounds, negative epigastric sounds, and a positive waveform capnography tracing. That last item is the one that carries the weight. Chest rise can be produced by gastric insufflation. Lung sounds transmit from a stomach in a noisy room. Waveform capnography is

measuring exhaled CO₂ with a shape to it, and CO₂ in that quantity with that morphology comes from alveoli. All four together is a properly confirmed airway, and it is documented as such.

Sizing deserves its own mention because it is where supraglottic airways quietly fail. The i-gel seats in the hypopharynx with its cuff conforming against the periglottic tissues, the bowl sitting over the laryngeal inlet and the tip in the upper esophagus. Too small and it sits high and leaks. Too large and it will not seat and it obstructs. There is no cuff to inflate to compensate for guessing wrong, which is a feature rather than a shortcoming: it forces the size decision to be correct up front. Theirs was, and the patient was being ventilated before ALS was on scene.

The second belongs to the arriving paramedic, who left it alone. Choosing not to intervene is a real clinical decision and it is harder to make than the alternative, because doing something feels like doing your job. Exchanging a working airway costs an interruption in ventilation, an interruption in compressions, and a nonzero chance of ending up with a worse airway than the one you started with. Nothing in this patient's presentation demanded a tube in that driveway. The airway was ventilating, capnography confirmed it, and the paramedic recognized that and moved on to the problems that were actually unsolved. That is good medicine and it should be named as such.

And another clinician could reasonably have exchanged it, and would also have been right. A supraglottic airway seals above the cords. It provides partial, not reliable, protection against aspiration. This patient had been vomiting before she arrested, and a clinician who weighed that and elected to place a cuffed tube below the cords would be making a defensible call on the same facts. The hospital did exactly that on arrival, in a controlled environment with suction, video, and a second set of hands, which is a very different risk calculation from the one on scene.

Both answers are correct. That is not a hedge, it is how clinical medicine actually works, and it is worth saying plainly to a group that spends most of its education being taught that there is one right answer and a protocol that contains it. There are questions with a single answer, and there are questions where two competent clinicians look at the same patient, weigh the same tradeoffs differently, and both practice well. Airway management generates a lot of the second kind. What separates good practice from bad is not which option you pick, it is whether you assessed, whether you confirmed, whether you can articulate why, and whether you kept watching afterward.

Everybody involved in this airway did that. The patient was ventilated early, correctly, and continuously, from first responder hands through to the emergency department, and that is the part of this call that went right.

Mechanical CPR

The large randomized trials comparing mechanical to manual compressions show equivalence, not superiority. What a device buys is consistency in the phases where human compressions degrade the most: fatigue after a few minutes, the move to the truck, and transport. What it costs is the pause to deploy it.

Here the device went on after manual CPR was already running, which is the correct order and worth noting. The failure mode with these devices is a crew arriving, reaching for the device first, and buying a long interruption in compressions to install a machine whose only advantage is that it does not get tired. This crew got hands on the chest first and the device on second.

Naloxone, intra-arrest

A competitive antagonist at the mu opioid receptor. It does not metabolize or eliminate the opioid; it occupies the receptor and displaces the agonist, reversing opioid-mediated respiratory depression within a couple of minutes when there is circulation to carry it there.

The suspicion behind it was reasonable and, as it turned out, aimed in the right direction: send-out toxicology on hospital day 1 confirmed hydrocodone, hydromorphone, and fentanyl. Naloxone is on the protocol and giving it was not wrong. It is still worth being blunt about what it could and could not do here, because this drug is reached for reflexively and the reflex outruns the pharmacology.

It cannot get to the receptor. In arrest there is no cardiac output to distribute it. Whatever goes into an IO sits in a circulation being moved only by compressions, at a fraction of normal output, and reaches its target slowly if at all.

The receptor is no longer the problem. By the time a patient is pulseless the opioid has already done its damage, which was apnea. The injury now in progress is global hypoxic-ischemic injury, and taking an opioid off a receptor does not put oxygen back into a brain. On top of that, the specific problem naloxone solves had already been solved mechanically: the airway was managed and the patient was being ventilated. The drug was being asked to duplicate work a BVM was already doing.

There is no outcome evidence for it in arrest. Resuscitation science does not support naloxone during cardiac arrest. The treatment for an opioid-associated arrest is the treatment for every other arrest: compressions, ventilation, and epinephrine.

And it can complicate the days that follow. This is the part that is invisible from the truck. Naloxone does not distinguish between the opioid that caused the arrest and the opioids a hospital is about to need. A patient who reaches an ICU requires sedation and analgesia to tolerate an endotracheal tube, and antagonist still bound at the receptor makes that harder, pushing the team toward higher doses or toward agents that carry their own hemodynamic cost in a patient who is already on pressors. In a patient with chronic opioid exposure, naloxone can also precipitate acute withdrawal: catecholamine surge, tachycardia, hypertension, agitation, and vomiting, in a myocardium that just arrested and an airway that already has gastric contents in it.

Result here: no discernible effect. ROSC followed the second milligram of epinephrine.

The honest summary is that naloxone in this arrest was defensible, unlikely to help, and not free. In a patient who is breathing badly it is among the highest-value drugs on the truck. In a patient who is pulseless it is not the drug that fixes the problem, and the decision worth examining is whether it displaced attention from the ones that do.

Epinephrine 1 mg IO, twice

At the doses used in arrest, the effect that matters is alpha-1 mediated vasoconstriction of peripheral arterioles. That raises aortic diastolic pressure during the recoil phase of compressions, which raises coronary perfusion pressure, which is the physiologic prerequisite for a heart to restart. The beta effects, increased heart rate and contractility, are largely a liability during arrest, because they raise myocardial oxygen consumption in a myocardium that has none.

Epinephrine has its best evidence in non-shockable arrest, which is what this was. What the evidence shows is a real increase in ROSC and in survival to 30 days, with no demonstrated improvement in favorable neurologic

outcome. That is worth sitting with, because it describes this case precisely. The drug did what it does. It produced ROSC in a patient whose brain had already been without oxygen too long. That is not an argument against giving it, it is an accurate description of what it can and cannot accomplish.

Fluid, 1,500 mL normal saline

Aimed at the relative hypovolemia and vasoplegia of post-arrest syndrome. It worked partially and briefly, and it had a cost that shows up in the labs. Normal saline is 154 mmol/L each of sodium and chloride, which is a substantially higher chloride concentration than plasma. Give enough of it and serum chloride rises. Because electroneutrality has to hold, bicarbonate falls to make room. The result is a **hyperchloremic non-anion-gap metabolic acidosis**, and this patient has a textbook example of it: chloride 108, bicarbonate 21, anion gap 8. That acidosis is iatrogenic and generally benign, but if you do not know to look for it, a falling bicarbonate looks like a patient getting worse.

Push dose epinephrine, 20 mcg IO, five doses

I want to emphasize: **this was very good care, and it is care that frequently does not happen.** Post-ROSC hypotension is common, it is lethal, and it is undertreated in the field. Plenty of crews watch a MAP in the low 50s after ROSC and give nothing but fluid, because push dose pressors mean drawing up a dilution under time pressure, committing to a decision without an order, and then staying with it. This crew recognized the hypotension on the first set, treated it, reassessed, and treated it again, five times across a transport, while also managing an airway and a fresh post-arrest patient. That is difficult to execute and it kept a brain perfused during the window when perfusion still counted for something.

Dilute epinephrine, 10 mcg/mL, given as intermittent boluses. Onset is under a minute; clinical duration is roughly five to ten minutes, because epinephrine is cleared rapidly by catechol-O-methyltransferase and monoamine oxidase.

Now read the vital signs sequence with that in mind. Pressure rises after a dose, then falls back before the next one. That pattern is not a failure of the drug or of the person pushing it. It is the pharmacokinetics doing exactly what they do, against a vasoplegia that does not resolve in ten minutes. Push dose pressors are a bridge, and what they bridge to is a continuous infusion. The emergency department reached the same conclusion shortly after arrival and started norepinephrine. The crew held the line until there was somebody to hand it to, which is precisely what the drug is for.

IN-HOSPITAL INTERVENTIONS

Endotracheal intubation

The i-gel was working and correctly confirmed, and the emergency department exchanged it anyway. Both of those things are fine, and the reason is environment more than physiology. In a patient who had been vomiting, with secretions already documented in the right mainstem, and who was going to need suctioning, controlled PEEP, and days of ventilation, a cuffed tube below the cords is the more definitive answer. It was placed on one video-assisted attempt with full confirmation. Making that exchange in a resuscitation bay with suction, a bougie, video, and a second airway operator standing there is a different proposition from making it on a floor.

The tube abutted the carina on CT and was withdrawn. This is worth a moment. A tube that deep will irritate the carina, and the depth changes with head position: neck flexion advances the tube toward the carina and right mainstem, extension withdraws it. Confirmation of placement is not a one-time event.

Mechanical ventilation

Assist control volume mode: the machine guarantees a set number of breaths at a set tidal volume, and delivers that same volume for any additional breath the patient triggers. PEEP holds a small positive pressure at end expiration, keeping alveoli from collapsing between breaths and improving oxygenation without requiring more FiO₂.

Tidal volume in critical care is set against **predicted body weight**, not actual weight, because lung size tracks height rather than adiposity. For a female of this patient's documented height, predicted body weight is approximately 52 kg, and her delivered volumes of 429 to 470 mL correspond to roughly 8.2 to 9.0 mL/kg predicted. Her FiO₂ was weaned from 40% to 30% over the course, which is the direction you want.

Norepinephrine, then epinephrine infusion

Norepinephrine is predominantly an alpha-1 agonist with modest beta-1 activity. It raises systemic vascular resistance with relatively little increase in heart rate, which is why it is first line in vasodilatory and undifferentiated shock. It targets exactly the lesion of post-arrest vasoplegia.

Epinephrine infusion was added on top, contributing beta-1 inotropy and chronotropy for the possibility of post-arrest myocardial stunning.

The stated target was a **MAP greater than 65**. That number is not arbitrary. Most vascular beds autoregulate flow across a range of pressures, and below roughly a MAP of 60 to 65 the autoregulatory reserve is exhausted and organ blood flow becomes pressure dependent. After cardiac arrest, cerebral autoregulation is impaired or absent, so the brain is pressure dependent even above that threshold. The number is a floor, not a goal.

Both were off within about a day, and the echocardiogram at 57% confirmed what that weaning implied: the myocardium was never the problem. This was distributive shock from the reperfusion response, and it resolved when the reperfusion response did.

Vancomycin and piperacillin-tazobactam, narrowed to ampicillin-sulbactam

Broad empiric coverage started for a plausible aspiration event in a patient in shock with a white count of 16.0. **Vancomycin** covers gram positive organisms including MRSA. **Piperacillin-tazobactam** covers gram negatives including pseudomonas plus anaerobes.

What is worth following here is how the coverage came back down. Vancomycin was stopped on a negative **MRSA nares PCR**, a swab of the anterior nares tested for the genetic material of methicillin-resistant staph aureus. The reason a negative nares swab can stop a systemic antibiotic is its negative predictive value: patients who do not carry MRSA in their nose very rarely have MRSA pneumonia, on the order of 95% or better. A negative test does not prove absence of infection, it makes one specific organism unlikely enough to stop covering for it. Piperacillin-tazobactam was then narrowed to **ampicillin-sulbactam**, which covers oral flora and anaerobes and gives up the anti-pseudomonal spectrum that a community aspiration does not need, with a defined five day stop date rather than an open-ended course.

Propofol, levetiracetam, fosphenytoin

Propofol potentiates GABA-A, the principal inhibitory receptor in the brain, producing rapid sedation and, at higher doses, burst suppression. Its two liabilities here are both relevant: it drops blood pressure by vasodilation and reduced sympathetic outflow, and it suppresses the neurologic exam. You cannot prognosticate a comatose patient who is on propofol, which is why the drug was discontinued and a second EEG recorded off it.

Levetiracetam binds synaptic vesicle protein 2A and reduces excitatory neurotransmitter release. It is used first in most seizure protocols because it is available intravenously, has minimal drug interactions, requires no level monitoring, and is hemodynamically neutral.

Fosphenytoin is a water-soluble prodrug converted to phenytoin, which blocks voltage-gated sodium channels in their inactivated state and prevents high-frequency repetitive firing. The prodrug exists because phenytoin itself is dissolved in propylene glycol at pH 12, which causes phlebitis and tissue injury and limits infusion rate. Fosphenytoin can be given faster and hurts less. It is dosed in phenytoin equivalents for exactly this reason.

What failed, and why it did not matter. All three drugs worked. The myoclonus resolved. And the second EEG, off propofol, showed the same burst suppression with identical bursts. That is the most important sentence in this case. **The seizures were a symptom of the injury, not the cause of it.** Post-arrest myoclonic status epilepticus is a marker of how much cortex died, and suppressing the electrical output of dead cortex does not bring it back. Treating it is still correct, because ongoing seizure activity adds metabolic demand and secondary injury, and because you cannot assess a brain that is seizing. But the treatment was never going to change the outcome, and understanding why is the difference between knowing a protocol and knowing a disease.

Imaging and echocardiography

CT head was looking for a hemorrhage or a mass as the cause of the arrest, and for the consequences of it. It found neither an acute cause nor, at that early hour, established anoxic edema. A CT head early after arrest is frequently normal even in a devastating injury, because cytotoxic edema takes hours to become visible. A normal early CT is not reassurance.

CT thorax, abdomen and pelvis with contrast was working the reversible causes with better tools than the field has: aortic dissection, central pulmonary embolism, tension pneumothorax, and, given the assault and fall history, occult traumatic injury. All negative. It also documented the bilateral rib fractures from CPR, which are an expected consequence of adequate compressions in a 71-year-old and not a complication to apologize for.

Echocardiogram was looking for pericardial tamponade, right ventricular strain suggesting pulmonary embolism, and regional wall motion abnormality suggesting occlusive coronary disease. EF 57% with no wall motion abnormality argues against all three, and against a primary cardiac cause for the arrest.

Comfort care and extubation

The endotracheal tube was removed because it no longer served a goal the family had chosen. Extubation to comfort measures is a deliberate intervention with its own physiology and its own medications, not the absence of treatment. She was transferred out of intensive care to inpatient hospice and died early the following day.

The labs, and what they mean

Reference ranges vary between institutions and between analyzers. The ranges below come from the reporting laboratory where the source report printed one. Where it did not, a standard adult range is supplied and is marked. Always use the range printed on the report in front of you.

Acid-base and lactate

Lab	Arrival (day 0)	Day 3	Reference range
pH	7.32 (venous)	7.44 (arterial)	Venous 7.32 to 7.43; arterial 7.35 to 7.45 (supplied)
PCO2 (mmHg)	39 (venous)	29 (arterial)	Venous 41 to 51; arterial 35 to 45 (supplied)
PO2 (mmHg)	36 (venous)	61 (arterial)	Venous 25 to 40; arterial 80 to 100 (supplied)
Bicarbonate (mmol/L)	21 (serum, day 1)	19.5 (arterial)	22 to 29
Base excess (mmol/L)	not drawn	-3.7	-2 to +2 (supplied)
O2 saturation (%)	64.8 (venous)	93 (arterial)	Venous 40.0 to 75.0; arterial 95 to 100 (supplied)
Anion gap (mmol/L)	8 (day 1)	not repeated	5 to 14
Chloride (mmol/L)	108 (day 1)	not repeated	98 to 107 (supplied)
Lactate (mmol/L)	1.9 (day 1 morning)	not repeated	0.0 to 2.0

pH measures hydrogen ion concentration. **PCO2** is the respiratory contribution to acid, controlled by ventilation. **Bicarbonate** and **base excess** are the metabolic contribution, controlled by the kidney and consumed by acid loads. **Anion gap** is sodium minus chloride and bicarbonate, and it estimates how much of a low bicarbonate is explained by unmeasured acids like lactate and ketones rather than by chloride.

Look at the day 3 gas. **The pH is 7.44 and every single component of it is abnormal.** PCO2 is 29, which is a respiratory alkalosis. Bicarbonate is 19.5 with a base excess of -3.7, which is a metabolic acidosis. The two errors are pointing in opposite directions and they happen to cancel. If you read only the pH, you would conclude this patient's acid-base status was normal. It was not. It was two abnormalities in balance.

The day 1 chemistry names the metabolic half. Bicarbonate 21 with an anion gap of **8** is a **non-anion-gap** acidosis. That is a critical distinction, because it means the acid is not lactate and not ketones. Chloride of 108 tells you what it is instead: 1,500 mL of normal saline in the field plus more in the department, delivering a chloride load that displaced bicarbonate. Hyperchloremic acidosis from resuscitation fluid.

Now the number that carries the most weight in this entire section: **lactate 1.9, which is normal.** By the morning after an hour of cardiac arrest, her lactate had already cleared. Lactate is produced when cells make ATP without oxygen and is cleared mostly by liver and kidney once perfusion returns. A normal lactate at that point is genuine and reassuring information about **circulation**. It tells you the shock was over and the organs were being perfused. **It tells you nothing whatsoever about the brain**, which had already sustained its injury during the arrest and would not un-sustain it because the plumbing was fixed. A patient with a perfect lactate can have a destroyed cortex.

Two more findings in the gas worth pulling out. The venous PCO₂ on arrival is 39, marked low, and it is low because a venous PCO₂ should exceed an arterial one; by the time this was drawn she was being ventilated. And on day 3, PO₂ of 61 on an FiO₂ of 0.30 gives a **P/F ratio of about 203**, which is a real oxygenation deficit consistent with the atelectasis and secretions on CT, sitting quietly underneath a normal-looking pH.

Heart: troponin and creatine kinase

Lab	Arrival	2 hr	Day 1	Reference range
hs troponin T (ng/L)	40	142	1,059	Less than 15 (female)
Delta from baseline (ng/L)	not applicable	102	not reported	Not applicable
Creatine kinase (U/L)	not drawn	not drawn	1,434	26 to 192 (supplied)
Ejection fraction (%)	not applicable	not applicable	57	55 to 70 (supplied)

Troponin is a regulatory protein of the cardiac contractile apparatus. It is not in the blood in meaningful quantity unless cardiac myocytes are damaged, and high-sensitivity assays detect it at very low concentrations, which is why the reference limit is 15 and not 150. **Creatine kinase** is an enzyme in all muscle, cardiac and skeletal, and is not cardiac specific.

Her troponin went 40, then 142, then 1,059. On its face that is a large myocardial infarction. It was not one. The 12 lead in the field showed sinus rhythm with no ST segment abnormality. The echocardiogram showed an ejection fraction of 57% with no wall motion abnormality. If a coronary artery had occluded and killed enough myocardium to release a troponin of 1,059, that territory would not be contracting normally on the echo and the ECG would very likely say something.

This is called **type 2 myocardial injury**. Type 1 is an occluded coronary artery, plaque rupture, a supply problem in one vessel. Type 2 is myocardial oxygen supply failing to meet demand for reasons that have nothing to do with a blocked artery. In this patient there were four simultaneous mechanisms: an hour of global no-flow ischemia, direct mechanical injury from chest compressions, high-dose exogenous catecholamine driving oxygen consumption in an ischemic myocardium, and prolonged hypotension after ROSC.

Get the direction of causation right. The arrest caused the troponin. The troponin did not cause the arrest. Every post-arrest patient will have a troponin that looks alarming, and reading it backwards leads to activating a cath lab for a patient who needs neuroprotection and pressors.

Creatine kinase of 1,434 is mildly elevated and is consistent with an hour on the floor, an hour of mechanical compressions, and myoclonic jerking. This is rhabdomyolysis in the mild sense, well short of the values that threaten kidneys.

Kidney, muscle, and urine

Lab	Value	Reference range
Creatinine (mg/dL)	0.58	0.50 to 1.00
eGFR (mL/min/1.73 m2)	Greater than 90	Greater than 60
BUN (mg/dL)	14	8 to 20 (supplied)
Urine blood (dipstick)	Small	Negative
Urine RBC (per hpf)	0	0 to 2 (supplied)
Urine protein	Trace	Negative
Urine glucose (mg/dL)	500	Negative
Blood glucose, field (mg/dL)	349	70 to 99
Serum glucose, day 1 (mg/dL)	122	70 to 99
Urine specific gravity	1.008	1.005 to 1.030 (supplied)
Urine WBC (per hpf)	1	0 to 5 (supplied)
Urine nitrite	Negative	Negative

Creatinine is a breakdown product of muscle creatine, produced at a rate proportional to muscle mass and cleared by glomerular filtration. **BUN** is the nitrogenous end product of protein metabolism. Together they estimate filtration.

She had an hour of cardiac arrest and her kidneys were untouched. Creatinine 0.58, eGFR above 90, no acute kidney injury at any point. Worth noting alongside that: a creatinine of 0.58 in an elderly woman of modest build is not necessarily as reassuring as it looks in isolation, because creatinine production scales with muscle mass, and a small elderly patient can have a substantially reduced filtration rate at a creatinine that reads mid-normal. Creatinine reports a balance between production and clearance, so when production is low the balance can look normal at a filtration rate that is not.

Now the pattern in the urinalysis. **The dipstick was positive for blood and the microscope found zero red cells.** The dipstick does not detect red cells. It detects the peroxidase activity of the **heme** group, and heme is present in hemoglobin and in myoglobin as well as inside intact red cells. Heme positive with no red cells means free heme pigment in the urine: hemolysis or myoglobinuria. With a creatine kinase of 1,434 from compressions, a fall, and myoclonus, this is **myoglobinuria**, and it is the urine confirming what the CK said. This is a two-second read once you know that the strip tests chemistry and the microscope tests cells, and that a disagreement between them is itself the finding.

Second unflagged item: **urine glucose of 500 with a serum glucose of 122.** The kidney reabsorbs essentially all filtered glucose until the serum concentration exceeds a threshold of roughly 180 mg/dL, above which the transporters saturate and glucose spills. A serum glucose of 122 does not spill. So the glycosuria is a record of what the serum glucose used to be, at the time the urine in the bladder was made. The crew wrote that number

down: **349**. The urine is a slower-moving witness than the blood, and it preserved the field finding for hours after the serum value had normalized.

And the number behind it is worth following to the end. **Hemoglobin A1c** measures the fraction of hemoglobin non-enzymatically glycosylated by circulating glucose. Because red cells live about 120 days, it is a weighted average of blood glucose over the preceding two to three months and cannot be moved by anything that happened this week. Hers was **5.7% (reference less than 5.7, supplied)**, corresponding to an estimated average glucose of 117. **She was not diabetic**. Her field glucose was 349.

This is called **stress hyperglycemia**. Catecholamines, endogenous from the arrest and exogenous from 2 mg of epinephrine plus five push doses plus two infusions, drive hepatic glycogenolysis and gluconeogenesis while inducing peripheral insulin resistance. Cortisol does the same on a slower clock. The result is a glucose in the 300s in a person with entirely normal glucose handling. The practical consequence: **a high field glucose in a critically ill patient is not a diagnosis**. It is a measure of how hard the stress response is running, which tracks how sick the patient is. By hospital day 3 her point of care glucose was 132, with insulin never having been a feature of her admission.

Liver enzymes and synthetic function

Lab	Value	Reference range
AST (U/L)	106	0 to 35
ALT (U/L)	36	0 to 35
Total bilirubin (mg/dL)	0.4	0.2 to 1.2 (supplied)
Alkaline phosphatase (U/L)	114	40 to 129 (supplied)
Albumin (g/dL)	3.0	3.5 to 5.2
Total protein (g/dL)	6.1	6.4 to 8.3
INR	1.1	0.8 to 1.2 (supplied)

AST and **ALT** are enzymes released when cells with high metabolic activity are injured. ALT is relatively liver specific. **AST is not**: it is abundant in cardiac muscle, skeletal muscle, and red cells as well as liver. **Bilirubin** is the product of heme breakdown and reflects the liver's conjugating and excretory function rather than cell injury. **INR** reflects the liver's synthetic function, since it makes most clotting factors.

Read this pattern rather than the individual numbers. AST is three times ALT. Bilirubin is completely normal. INR is normal.

Cell membranes leaked, and the liver's actual jobs, conjugating bilirubin and manufacturing clotting factors, were being done normally. This is **not hepatocellular disease**. Some of that AST is not even liver: with a CK of 1,434 from compressions and myoclonus, skeletal muscle is contributing. What remains is a mild ischemic hepatitis, the beginning of what would be called **shock liver** if it had gone further.

It is worth stating what did not happen. Shock liver after a prolonged arrest classically produces transaminases in the hundreds to thousands within 24 to 48 hours, and it did not happen here. Perfusion was restored fast enough

and sustained well enough that the liver escaped with a scratch. That is the same story the creatinine tells and the same story the ejection fraction tells. **Everything downstream of the heart recovered. Only the brain did not.**

Albumin of 3.0 and total protein of 6.1 are partly dilution from resuscitation fluid and partly the acute phase response, in which the liver shifts synthesis away from albumin toward inflammatory proteins. That low albumin matters for the next section.

Electrolytes, calcium, and osmolality

Lab	Value	Reference range
Sodium (mmol/L)	137	135 to 145 (supplied)
Potassium (mmol/L)	3.5	3.5 to 4.5
Magnesium (mg/dL)	2.0	1.7 to 2.2 (supplied)
Calcium, total (mg/dL)	7.8	8.6 to 10.4
Calcium, ionized (mmol/L)	1.15	1.15 to 1.33
Osmolality, measured (mOsm/kg)	302	275 to 295 (supplied)
Methanol (mg/dL)	Less than 5	Not detected
Acetone (mg/dL)	Less than 5	Not detected
Ethanol (mg/dL)	Less than 10	Not detected

Sodium, potassium, and magnesium were all normal, which is itself informative: electrolyte derangement was not the cause of this arrest.

The calcium is worth working through. Total calcium was flagged low at 7.8. Ionized calcium was 1.15, which is normal. These do not conflict.

About 40% of circulating calcium is bound to albumin and is physiologically inert. Only the ionized fraction does anything: it drives cardiac and skeletal muscle contraction, neurotransmitter release, and the clotting cascade. **A total calcium is a measurement of the active hormone plus its inactive carrier.** When albumin falls, the bound fraction falls with it, the total drops, and nothing has actually happened to the patient. The rough correction adds 0.8 mg/dL of calcium for every 1.0 g/dL the albumin sits below 4.0. Her albumin is 3.0, so her corrected total calcium is about 8.6, which is normal, and the directly measured ionized value agrees. **Chasing a low total calcium in a hypoalbuminemic patient means treating an arithmetic artifact.**

One more worth working through. Measured osmolality was 302. Calculated osmolality, twice the sodium plus glucose over 18 plus BUN over 2.8, comes to about 286. The difference is an **osmolar gap** of roughly 16, where normal is under about 10. An osmolar gap means unmeasured osmotically active particles in the serum, and the classic reason to care is toxic alcohol ingestion. That is precisely why methanol, acetone, and ethanol were sent, and all three were negative. A gap with negative alcohols has other explanations, including infused drug carriers such as the lipid emulsion propofol is suspended in, which she was receiving. The habit is what matters here rather than the result: **when the measured osmolality exceeds the calculated one, something is in the blood that you did not measure, and toxic alcohols are the thing you rule out first.**

White count and differential

Lab	Arrival	Approx 9 hr	Approx 14 hr	Reference range
White blood cell ($\times 10^9/L$)	approx 23.6 (from differential)	16.0	approx 6 (from differential)	4.0 to 10.8
Neutrophils, absolute ($\times 10^9/L$)	15.34	not reported	3.17	1.8 to 7.7 (supplied)
Lymphocytes, absolute ($\times 10^9/L$)	6.84	not reported	1.97	1.0 to 4.8 (supplied)
Monocytes, absolute ($\times 10^9/L$)	1.18	not reported	0.58	0.2 to 0.9 (supplied)
Immature granulocytes (%)	not reported	not reported	0.5	Less than 1.0 (supplied)

Some vocabulary first, because it is used loosely and the terms carry real information. **Leukocyte** and white blood cell mean the same thing. The suffix tells you direction: **leukocytosis** is too many, **leukopenia** too few, and the same pattern applies to individual lines, so neutrophilia and neutropenia, lymphocytosis and lymphopenia. The **white blood cell count** is the total of all of them in a given volume of blood. The **differential** breaks that total into the individual cell types, and each type does a different job.

- **Neutrophils** are the front line against bacteria. They are the most numerous white cell, they engulf and destroy organisms directly, and they live only hours to a couple of days, so the marrow produces them continuously. When bacterial infection demands more than the marrow has finished, it releases immature forms early. Those show up on the report as **bands** or **immature granulocytes**, and immature cells in the blood mean the marrow is being pushed hard.
- **Lymphocytes** run the adaptive immune system: B cells make antibodies, T cells kill infected cells and coordinate the rest of the response. They are the dominant responders to viral infection. They are also long-lived, which means their numbers do not swing quickly in response to bacteria.
- **Monocytes** are cleanup and reconnaissance. They leave the bloodstream and become macrophages in tissue, where they engulf debris and dead cells and present fragments of pathogens to lymphocytes.
- **Eosinophils** and **basophils** are small populations involved in parasitic infection and allergic response.

The differential is reported two ways and the difference matters. The **percentage** is the share of the total that each type represents. The **absolute count** is that percentage multiplied by the total white count, and it is the number to read. A neutrophil percentage of 52 sounds unremarkable, but 52% of a very high total is a very high number of neutrophils, and 52% of a very low total is not enough to fight anything. Percentages describe proportion. Absolutes describe how many cells the patient actually has.

Follow the trend across the three columns, because the trend is the finding. On arrival the absolute counts sum to roughly 23.6. About nine hours later the total was 16.0. About five hours after that it was approximately 6. **The white count normalized in well under a day, and no antibiotic works that fast.**

Now look at which cells were up. Neutrophils were elevated at 15.34, which is what people expect. But **lymphocytes were also elevated at 6.84, and monocytes at 1.18**. Every line rose together, and that is the signature.

This is called **demargination**, and it is not infection. At any moment a large fraction of circulating leukocytes are not really circulating, they are rolling along vessel walls and sequestered in the spleen, lungs, and marginal pools. Catecholamines acting on beta-2 receptors reduce the adhesion that holds them there and flush them into the measured circulation within minutes. Nothing was produced. Cells that were already in the body moved into the compartment the sample was drawn from.

The discriminator is worth memorizing. **Bacterial infection produces neutrophilia with lymphopenia**, often with immature granulocytes and band forms as the marrow ramps up production. **Acute catecholamine stress produces elevation across all lines**, including lymphocytes, and it resolves in hours. This patient had endogenous catecholamines from the arrest plus 2 mg of epinephrine plus five push doses plus infusions of norepinephrine and epinephrine. Her immature granulocyte fraction was 0.5%, which is normal. Her blood cultures showed no growth.

This is why the record describes the leukocytosis as reactive versus aspiration-related and never resolved it, and it is why the antibiotics were narrowed rather than escalated. A white count of 16 in a post-arrest patient is closer to a catecholamine level than to an infection.

Red cells, morphology, and coagulation

Lab	Value	Reference range
Hemoglobin (g/dL)	10.0	12.0 to 16.0
Hematocrit (%)	30.5	37.0 to 47.0
MCV (fL)	94.4	80.0 to 100.0
MCHC (g/dL)	32.8	32.0 to 37.0
RDW (%)	14.6	11.5 to 14.5 (supplied)
Platelet (x10 ⁹ /L)	250	150 to 400 (supplied)
Prothrombin time (s)	11.6	11.0 to 13.5 (supplied)
INR	1.1	0.8 to 1.2 (supplied)
aPTT (s)	24.2	25 to 35 (supplied)
Echinocytes	Many	None
Schistocytes	Few	None

Hemoglobin 10.0 with a normal MCV is a **normocytic** anemia. No baseline is available, and at least part of it is dilutional: 1,500 mL of saline in the field plus more in the department expands plasma volume without adding red cells, lowering the concentration without changing the mass.

The morphology line is where this panel sets a trap. **Schistocytes** are fragmented red cells, physically sheared as they are forced past fibrin strands or damaged vessel lining. The reflex on seeing them is **thrombotic microangiopathy**, a group of conditions in which small vessels fill with clot and grind up the red cells passing through. Two carry that label most often.

Disseminated intravascular coagulation (DIC) is the clotting cascade switching on throughout the body at once, usually triggered by sepsis, major trauma, or prolonged shock. Clotting factors and platelets get consumed forming microclots in small vessels everywhere until there are not enough left, so the patient is simultaneously clotting and bleeding.

Thrombotic thrombocytopenic purpura (TTP) is a deficiency of the enzyme that normally trims von Willebrand factor down to working size. The untrimmed strands are abnormally sticky, platelets pile onto them inside small vessels, and the resulting plugs both shear red cells and starve the organ downstream.

That reflex is usually correct and it is wrong here, for a reason sitting in the next three rows of the same panel. Both of those conditions consume things. Platelets fall, often steeply, and in DIC the INR and aPTT prolong as clotting factors are used up. **Her platelets were 250, her INR was 1.1, her aPTT was 24.2.** Nothing was being consumed, which means nothing was being formed in bulk, which means the fragments have some other and far less interesting explanation. **Echinocytes**, or burr cells, are most often an artifact of specimen handling or storage time rather than a disease at all.

The larger point: **morphology is a prompt, not a diagnosis.** The value of noticing schistocytes is that it sends you to the platelets and the coagulation panel. When those are clean, the question is answered.

Toxicology

Test	Screen, day 0	Confirmatory, day 1	Reference
Opiate 300 (morphine class)	Negative	Morphine and codeine not detected	Not detected
Hydrocodone	Not on screen panel	389 ng/mL, detected	Not detected
Hydromorphone	Not on screen panel	738 ng/mL, detected	Not detected
Oxycodone	Negative	Not detected	Not detected
Fentanyl and metabolite	Negative	Both detected, low level	Not detected
Cannabinoids	Presumptive positive	THC metabolite greater than 200 ng/mL	Not detected
Benzodiazepine, cocaine, amphetamine, methadone, barbiturate	Negative	Not detected	Not detected
Acetaminophen (mcg/mL)	Less than 5.0	Not repeated	Therapeutic 10 to 30 (supplied)
Ethanol (mg/dL)	Less than 10	Less than 20	Less than 10 (supplied)

Read the opiate row and the hydrocodone row together. **The screen said no opiates. Confirmatory testing found hydrocodone and hydromorphone.**

That is not a lab error. A drug **screen** is an immunoassay: antibodies raised against a target molecule, which bind that molecule and anything shaped closely enough like it, above a fixed cutoff. The standard opiate assay is built around morphine and codeine, and semisynthetic opioids have modified structures that cross-react unpredictably with it. Hydrocodone and hydromorphone often fail to trigger it at ordinary therapeutic concentrations. Oxycodone and fentanyl cross-react so poorly they require their own dedicated assays, which is why they appear as separate lines rather than under opiates. The **confirmatory** test, resulting about a day later, identifies compounds by molecular mass instead of by shape, so it names the specific drug.

The portable version: **a negative drug screen is a statement about a panel of antibodies, not about a patient.** It rules out the compounds the assay was built for, above a cutoff, in urine. It rules out nothing else.

Hydrocodone with its metabolite hydromorphone is consistent with the medication dispensed to her the day before, though urine concentrations do not quantify a dose and cannot be worked backward to one. The fentanyl is not accounted for by anything in the record.

The chart never assigned an etiology to this arrest, and neither does this document. What is documented is the sequence: an older adult, opioid naive enough to carry an oxycodone intolerance on file, started a new opioid, developed nausea and vomiting, then respiratory distress, then a hypoxic PEA arrest, inside of a day.

Infection workup

Test	Result	Reference
Blood cultures, peripheral	No growth	No growth
MRSA nares PCR	Not detected	Not detected
Influenza A and B	Negative	Negative
SARS-CoV-2 PCR	Not detected	Not detected
Hepatitis C antibody	Non-reactive	Non-reactive
Urine nitrite	Negative	Negative
Urine leukocyte esterase	Negative	Negative

Uniformly negative, and that matters as much as a positive would. It is what allowed vancomycin to stop, piperacillin-tazobactam to narrow, and a five day stop date to be set instead of an open course. Negative tests are not wasted tests. They are what makes it safe to take treatment away.

Learning points

1. **A patient who is hypoventilating is on a shorter clock than a patient who is hypotensive.** The body stores no oxygen. Roughly a liter and a half exists in the whole system against a consumption of 250 mL a minute, which is why respiratory failure becomes cardiac arrest in minutes while shock takes hours. In this patient the

interval that decided the outcome was before the arrest, not after it. Any call where the complaint is vague but the respiratory rate or depth is wrong is a call where the highest-value intervention is ventilation, and it is available in every truck at every level.

2. **PEA is a diagnosis to make, not a rhythm to treat.** Organized electrical activity without a pulse means the conduction system outlived the pump, and the reason it outlived the pump is the actual disease. Compressions and epinephrine buy time; they do not fix hypoxia, hypovolemia, tamponade, tension, or toxin. In this case the cause was hypoxia, and the CT scanner, the echocardiogram, and the toxicology panel were the hospital running the same differential the crew ran, with better instruments and a day to do it.
3. **A negative drug screen is a statement about a panel of antibodies, not about a patient.** Immunoassays detect the compounds they were built for, above a fixed cutoff. Semisynthetics cross-react unpredictably, and synthetics often need their own dedicated assay or are invisible entirely. This patient's opiate screen was negative while confirmatory testing found hydrocodone, hydromorphone, and fentanyl. When history and physiology say opioid and the screen says no, believe the physiology. That principle transfers directly to the alert patient with pinpoint pupils whose chart says the screen was clean.
4. **After an arrest, an alarming number usually reports the arrest rather than explaining it.** A troponin of 1,059 with a normal ECG and a normal ejection fraction is type 2 injury from global ischemia, compressions, and catecholamines. A white count of 23 that falls to normal within a day with lymphocytes elevated alongside neutrophils is catecholamine demargination, not sepsis. A glucose of 349 with an A1c of 5.7 is stress hyperglycemia, not diabetes. In every one of these, the resuscitation produced the number. Ask what the number would look like if this patient had no disease other than having just been resuscitated, and treat what is left over.
5. **Restoring circulation is not restoring the patient, and the labs will not tell you which one you got.** By the morning after an hour of arrest this patient had a normal lactate, a normal creatinine, an ejection fraction of 57%, a normal INR, and by the following day a pH of 7.44. Every one of those is real. None of them says anything about the brain, which had already taken its injury during the arrest and which sets the ceiling on the outcome regardless of how well everything downstream recovers. The variables that determine that ceiling are recognition, ventilation, compression quality, and the minutes between collapse and flow. Those are field variables, and after ROSC nobody gets them back.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>